

HumanDiffusion: A Vision-Based Diffusion Trajectory Planner with Human-Conditioned Goals for Search and Rescue UAV

Faryal Batool, Iana Zhura*, Valerii Serpiva*, Roohan Ahmed Khan*, Ivan Valuev, Issatay Tokmurziyev, and Dzmitry Tsetserukou

Intelligent Space Robotics Laboratory,
Skolkovo Institute of Science and Technology, Skoltech

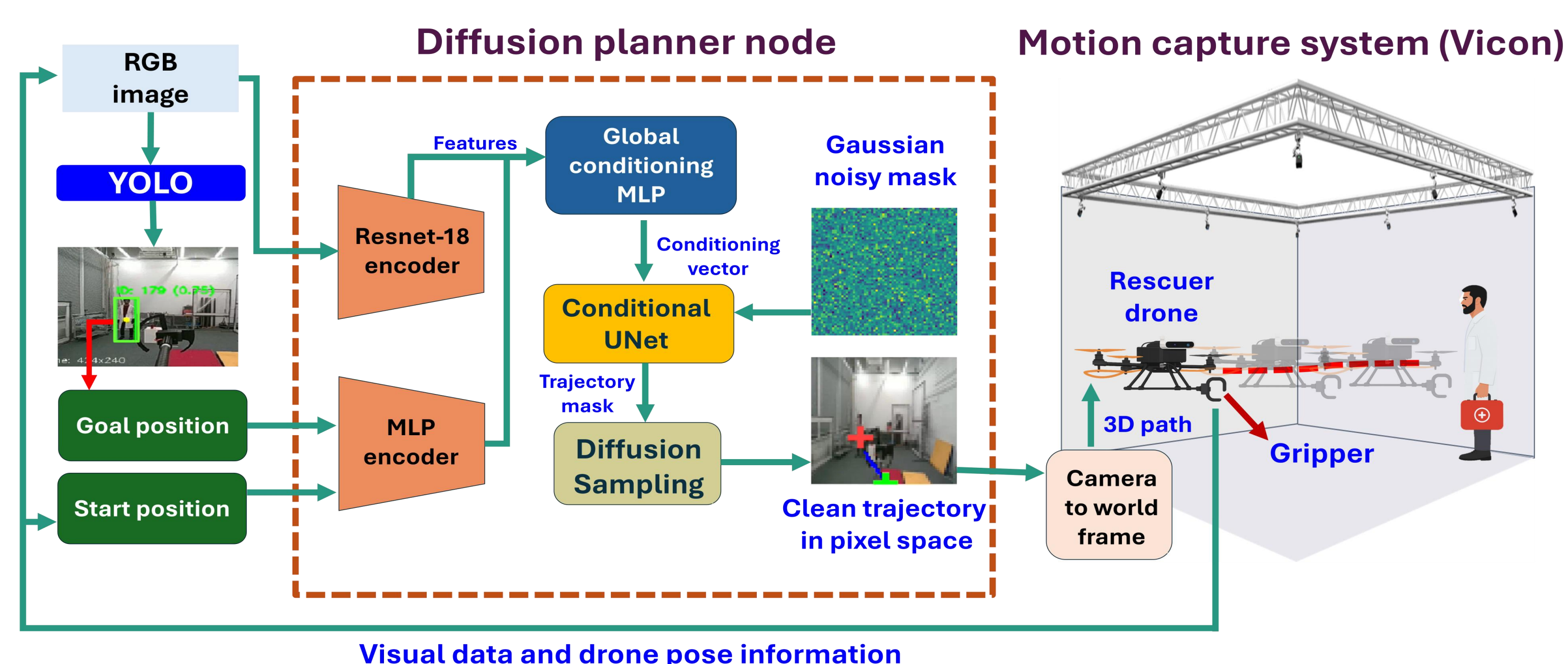


MOTIVATION

Autonomous UAV navigation in **search-and-rescue** environments requires detecting humans, **inferring navigation goals**, and operating safely in dynamic conditions. However, existing UAV navigation pipelines exhibit several key limitations:

- Dependence on **predefined goals or waypoints**.
- Reliance on **explicit maps or occupancy grids**.
- Dependence on classical planners (A*, RRT*, MPC) with strong **environment assumptions**.
- Limited coupling between human detection and global trajectory planning.

SYSTEM ARCHITECTURE



YOLO detects humans and provides goal points. The RGB image and start-goal information are encoded and fused to condition a UNet-based diffusion model. The model generates a clean pixel-space trajectory, which is projected into a 3D world-frame path for execution by UAV with a gripper.

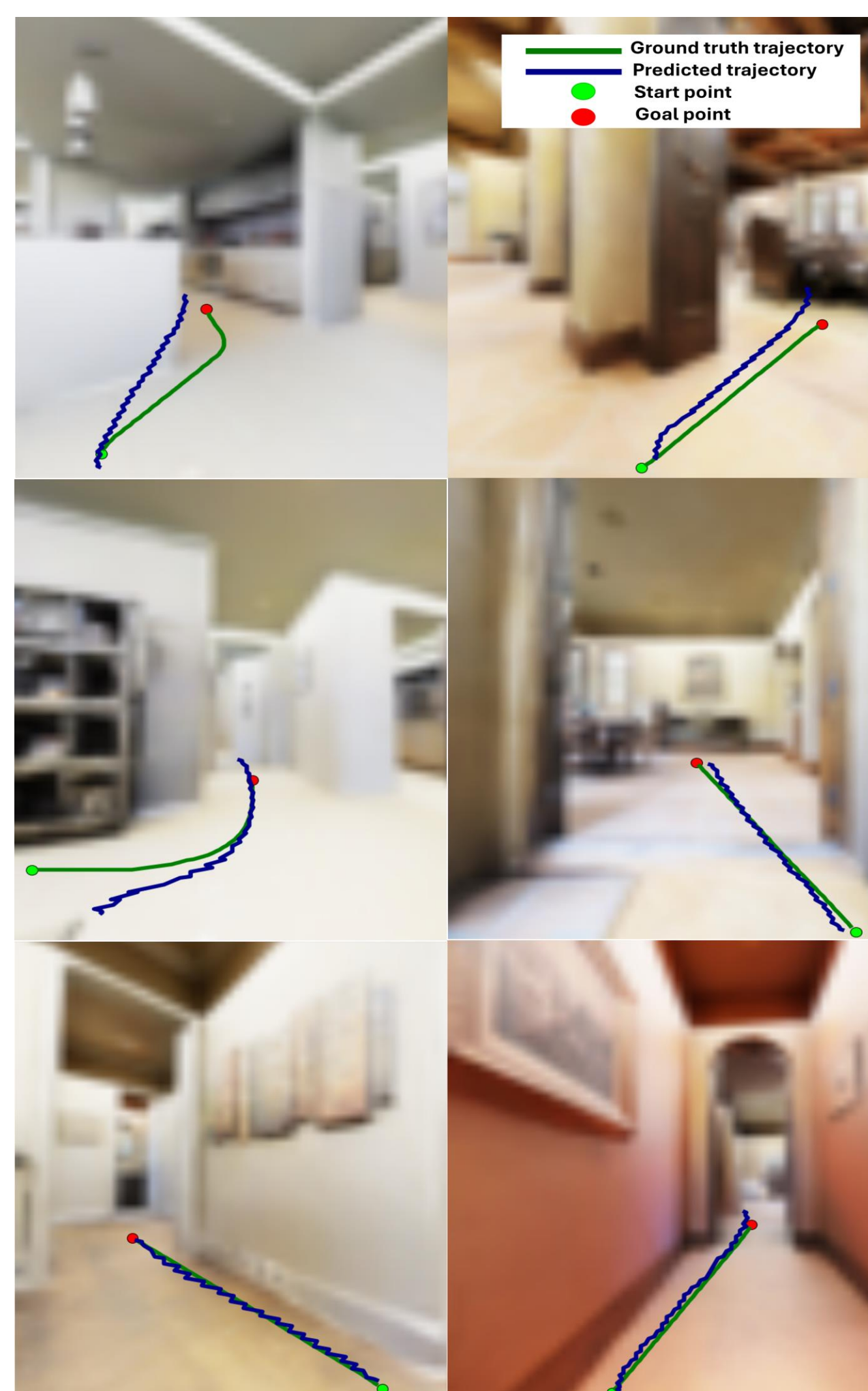
PROPOSED SOLUTION

HumanDiffusion is a vision-based diffusion planner enabling human-aware UAV navigation in search-and-rescue scenarios. Our key contributions are:

- **Human-Conditioned Goal Inference:** Uses YOLO-based human detection to automatically infer navigation goals from bounding boxes.
- **Image-Conditioned Diffusion Planning:** Generates smooth global trajectories directly from RGB images and start goal information.
- **Sim-to-Real Deployment:** The diffusion model is **trained entirely on simulated images** and **deployed zero-shot in real-world UAV experiments**.
- **Map-Free Navigation:** Enables perception-driven UAV navigation without explicit maps or predefined waypoints.

EXPERIMENTAL RESULTS

Evaluation on a Simulated Test Dataset

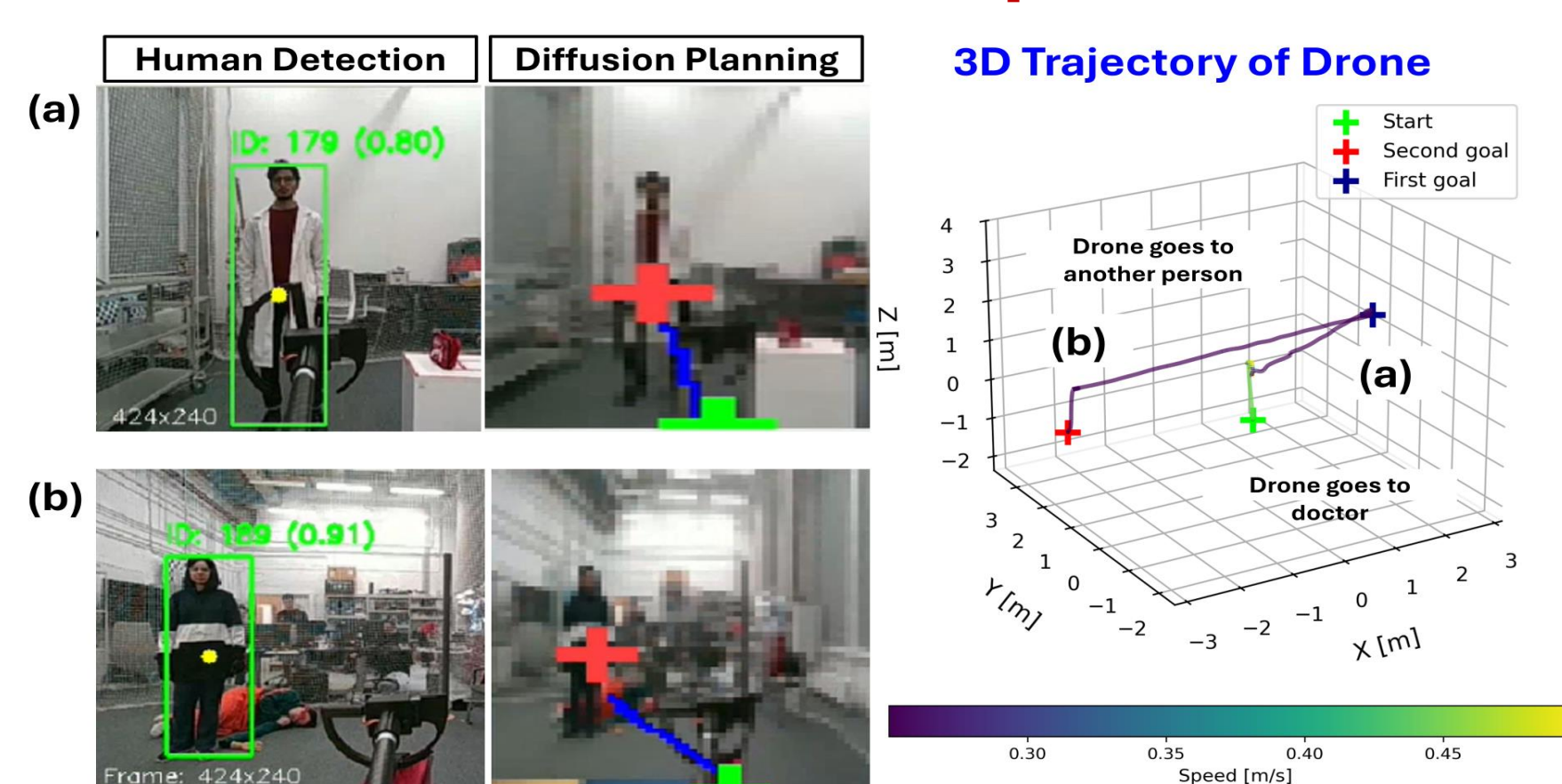


Metric	Value
Trajectory RMSE	0.14 pixels
Start/Goal RMSE	0.045 pixels
Trajectory IoU	0.49

The diffusion planner achieves a **trajectory error of 0.14 pixels Root Mean Squared Error (RMSE)** with an Intersection over Union (IoU) of **0.49**, indicating strong agreement between predicted and ground-truth trajectory masks. Start and goal prediction errors remain very low (**≈0.045 pixels RMSE**), demonstrating accurate boundary condition estimation.

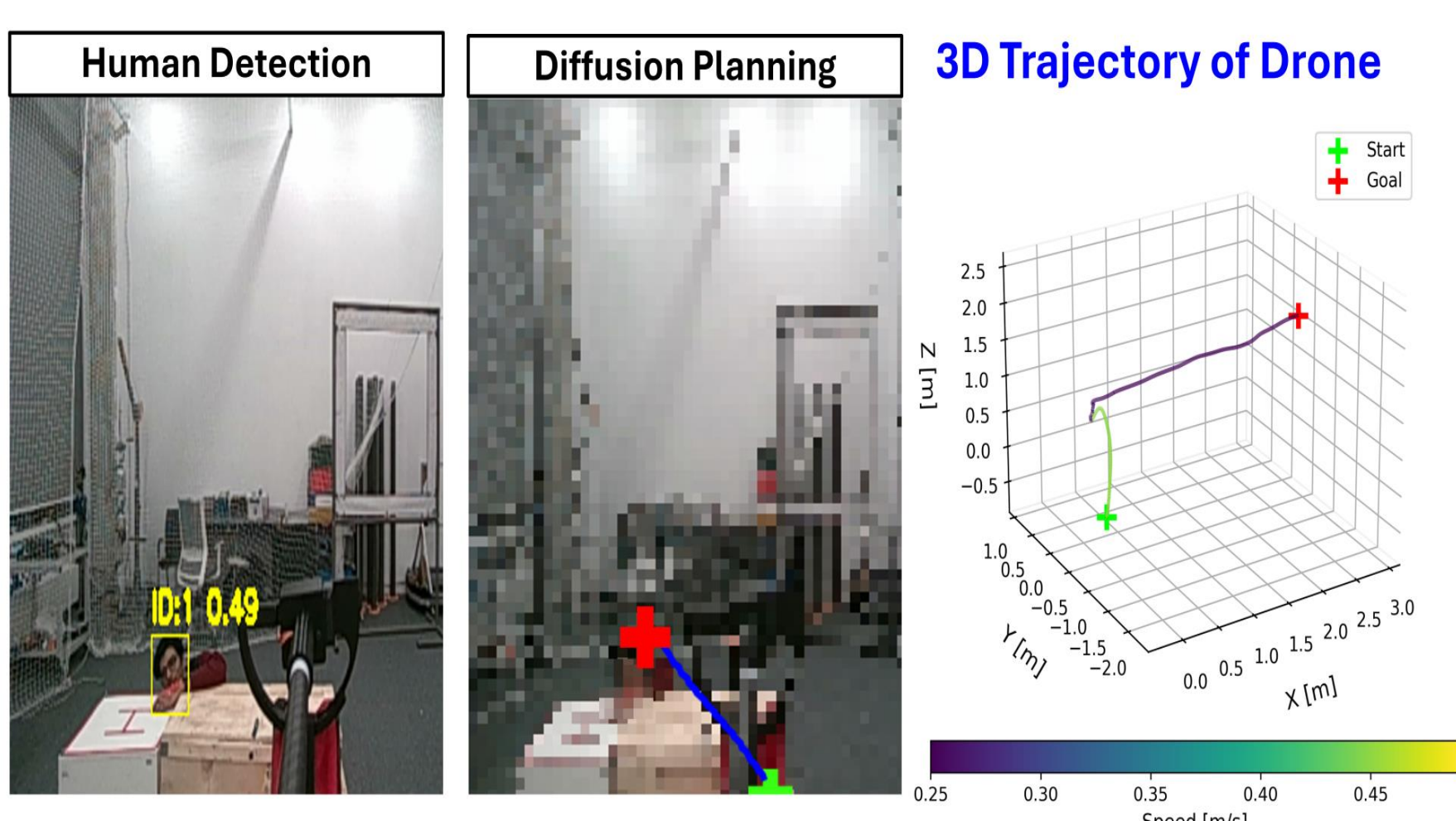
Evaluation on Real Indoor Scenarios

Scenario 1: Accident Response



Across **10 trials**, the UAV successfully completed **9 full delivery cycles** while maintaining a safe stopping distance of **1 m** from participants.

Scenario 2: Search-and-Locate in Occluded Environments



The system succeeded in **7 out of 10 trials**. Performance was affected by communication delays and limited arena size.

Scenario	Success Rate
Accident Response	90 %
Occluded Search	70 %
Overall	80 %

CONCLUSION

- **HumanDiffusion enables map-free UAV navigation** by combining human detection with diffusion-based trajectory generation from RGB images.
- **The model is trained entirely on simulated images and deployed zero-shot in the real world**, achieving an overall **80% mission success rate** in accident-response and search-and-locate scenarios.

